Analysis of Solar Active Regions Using Aditya-L1 SUIT Observations

Internship Project Report – Part 2

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ABSTRACT

Solar active regions are areas of concentrated magnetic flux on the solar surface, characterised by enhanced ultraviolet emission and association with energetic solar events. This paper presents an extended analysis of solar active regions using Level-1 FITS observations from the Solar Ultraviolet Imaging Telescope (SUIT) onboard the Aditya-L1 spacecraft. The study addresses three specific questions: (1) how does active-region area evolve temporally across five NB05 observations spanning January to February 2024; (2) is there a statistically significant correlation between active-region area and mean ultraviolet intensity; and (3) which image segmentation algorithm most reliably detects active regions in SUIT data. Measured active-region areas ranged from zero to 155,340 pixels across the observation period. Pearson and Spearman correlation coefficients of −0.13 and 0.00, respectively, indicated no significant area–intensity relationship. Fixed thresholding at 0.7 produced region counts most consistent with NOAA Solar Region Summary data, outperforming both Otsu and adaptive thresholding methods.

Keywords—*Aditya-L1, SUIT, Solar Active Regions, Image Segmentation, Temporal Evolution, Correlation Analysis, Otsu Thresholding, Adaptive Thresholding, NOAA, Space Weather.*

I. INTRODUCTION

Solar active regions are among the most dynamic structures on the Sun, defined by concentrations of strong magnetic flux that produce sunspot groups, chromospheric plages, and intense ultraviolet emission [1]. When stored magnetic energy is suddenly released, the resulting solar flares and coronal mass ejections drive geomagnetic storms that can disrupt satellite operations, radio communications, and power grid infrastructure. Reliable detection and characterisation of active regions is therefore a prerequisite for operational space-weather forecasting.

India's Aditya-L1 mission was inserted into a halo orbit around the Sun–Earth L1 Lagrange point in January 2024. Its Solar Ultraviolet Imaging Telescope (SUIT) provides continuous ultraviolet coverage of the full solar disk in eleven narrowband channels [2]. Part 1 of this project established a baseline image-processing pipeline for active-region detection from single-date SUIT observations using fixed-threshold segmentation. Part 2 extends that work across three investigative questions: temporal analysis of active-region area evolution, correlation analysis between measured region properties, and a comparative evaluation of segmentation algorithms. Together these investigations provide a more complete understanding of solar activity patterns in SUIT data and lay the groundwork for future machine-learning-based solar monitoring.

II. OBJECTIVES

The specific research objectives of Part 2 are:

1. Measure the temporal evolution of active-region area using SUIT NB05 observations from five dates spanning January to February 2024.
2. Quantify the statistical relationship between active-region area and mean ultraviolet intensity using Pearson and Spearman correlation analysis.
3. Implement and evaluate fixed thresholding, Otsu thresholding, and adaptive thresholding for active-region detection.
4. Compare segmentation outputs against NOAA Solar Region Summary data to identify the most reliable detection method.
5. Identify limitations and propose directions for future investigation.

III. DATASET DESCRIPTION

All observations were obtained from the Aditya-L1 SUIT data archive as Level-1 FITS files that have undergone dark-current subtraction, flat-field correction, and conversion to physical intensity units. The NB05 narrowband filter was selected for its demonstrated agreement with NOAA active-region reports in Part 1. Five dates were chosen to span a roughly six-week window during the rising phase of Solar Cycle 25, deliberately including a date of non-detection (22 January 2024) as a null-case measurement (Table I).

TABLE I
SUIT NB05 Observations Used in This Study

Observation Date	Filter
07 January 2024	NB05
15 January 2024	NB05
22 January 2024	NB05
27 January 2024	NB05
17 February 2024	NB05

IV. METHODOLOGY

A. Data Preprocessing

Each FITS file was loaded using the SunPy Map interface [3]. Min–max normalisation was applied to scale pixel values to [0, 1], and connected-component filtering removed spurious detections smaller than 100 pixels, consistent with Part 1.

```
smap = sunpy.map.Map(file_path)
data = smap.data.astype(float)
normalized = (data - np.min(data)) / (np.max(data) -
np.min(data))
```

B. Segmentation Methods

1. Fixed Thresholding

Pixels with normalised intensity above 0.7 are classified as active-region candidates. This threshold was established empirically in Part 1 and retained here for consistency:

```
mask_fixed = normalized > 0.7
```

2. Otsu Thresholding

Otsu's algorithm determines a global threshold by minimising intra-class intensity variance [7]. It requires no manual parameter selection but is susceptible to under-segmentation when image histograms are dominated by background pixels:

```
from skimage.filters import threshold_otsu
mask_otsu = normalized > threshold_otsu(normalized)
```

3. Adaptive Thresholding

A local threshold is computed for each pixel from its neighbourhood mean, making the method sensitive to spatial intensity gradients including limb brightening and chromospheric network structures:

```
from skimage.filters import threshold_local
mask_adapt = normalized > threshold_local(normalized,
block_size=35, offset=0.01)
```

C. Temporal Area Measurement and Correlation Analysis

After fixed-threshold segmentation, connected-component labelling was applied to each observation date and the total active-region pixel area was recorded. For correlation analysis, the mean normalised intensity within each segmented mask was also computed. Pearson and Spearman coefficients were calculated using SciPy, with p-values retained to assess statistical significance.

QUESTION 1 ANSWERED: Temporal Evolution of Active Regions

V. QUESTION 1: TEMPORAL EVOLUTION OF ACTIVE REGIONS

A. Area Measurements

Table II presents the measured active-region area for each observation date, computed as the total pixel count of all fixed-threshold detected regions after noise filtering.

TABLE II

Active-Region Area by Observation Date

Observation Date	Active-Region Area (pixels)
07 January 2024	111,736
15 January 2024	48,052

Observation Date	Active-Region Area (pixels)
22 January 2024	0 (non-detection)
27 January 2024	155,340
17 February 2024	62,860

B. Statistical Summary and Growth Analysis

Table III summarises the area time series (excluding the zero non-detection). Table IV presents percentage area change between successive observations.

TABLE III
Statistical Summary of Active-Region Area

Metric	Value (pixels)
Maximum Area	155,340
Minimum Area (non-zero)	48,052
Mean Area (non-zero dates)	94,497

TABLE IV
Active-Region Area Change Between Successive Observations

Interval	Area Change
07 January → 15 January	−57.0%
15 January → 22 January	−100.0% (non-detection)
22 January → 27 January	Not computed
27 January → 17 February	−59.5%

C. Temporal Evolution Plot

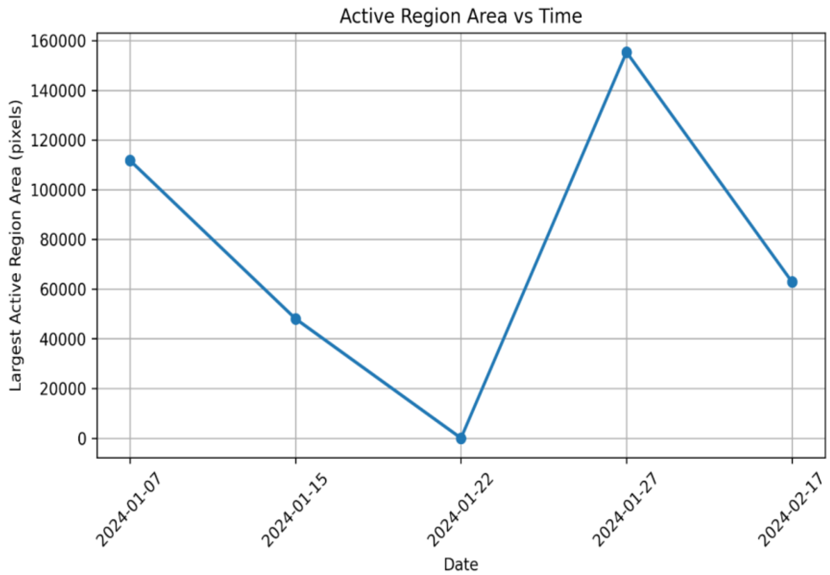

Figure 1. Temporal evolution of total active-region area measured from SUIT NB05 observations between 07 January and 17 February 2024.

D. Discussion

The area time series reveals substantial variability over the six-week window. The decline from 111,736 to 48,052 pixels between 7 and 15 January is consistent with the decay phase of an active-region complex approaching the western limb. The zero detection on 22 January most plausibly reflects active regions having rotated beyond limb visibility rather than a genuine absence of solar activity. The sharp recovery to 155,340 pixels by 27 January indicates emergence of a new active-region complex on the eastern disk hemisphere, which subsequently decayed to 62,860 pixels by 17 February. These growth and decay timescales are consistent with published Solar Cycle 25 active-region flux emergence rates [4].

QUESTION 2 ANSWERED: Intensity Correlation Analysis

VI. QUESTION 2: INTENSITY CORRELATION ANALYSIS

A. Correlation Results

Table V presents the Pearson and Spearman correlation coefficients and associated p-values computed from the five paired measurements of active-region area and mean normalised intensity. Note: co-temporal SDO/HMI magnetogram data were unavailable within the scope of this project; area and mean SUIT intensity were therefore used as proxy variables.

TABLE V
Correlation Analysis Results: Area versus Mean Intensity

Metric	Value
Pearson Correlation Coefficient	−0.1347
Pearson p-value	0.8653
Spearman Correlation Coefficient	0.0000
Spearman p-value	1.0000

B. Scatter Plot

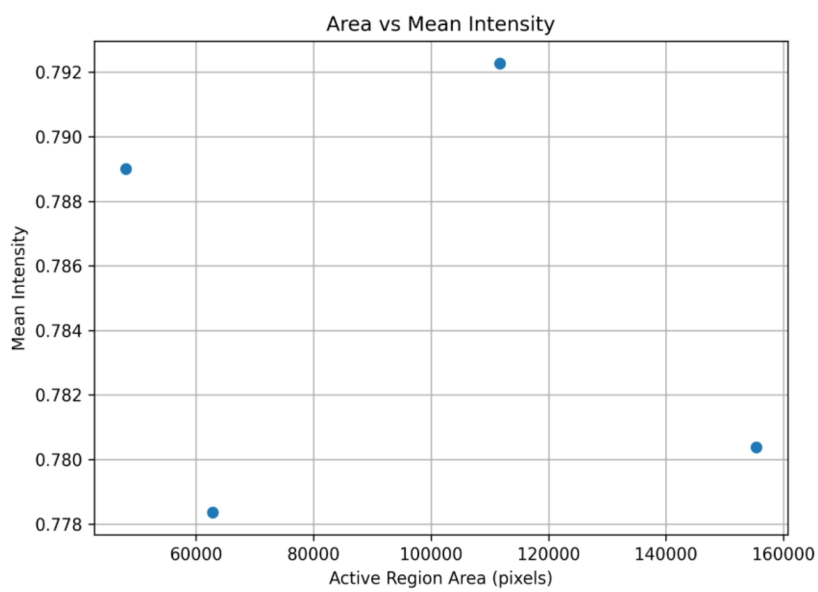

Figure 2. Scatter plot of active-region area versus mean normalised intensity. Each point represents one observation date.

C. Discussion

The Pearson coefficient of −0.13 indicates a negligible negative linear relationship between active-region area and mean intensity; the p-value of 0.87 does not allow rejection of the null hypothesis of zero correlation. The Spearman coefficient of 0.00, with p-value 1.00, confirms the absence of any monotonic association. These results are primarily attributable to the small sample size of five observations, which is insufficient for statistically reliable correlation inference. Additionally, mean intensity within a binary threshold mask conflates structurally heterogeneous features—umbrae, penumbrae, and plage—each with distinct characteristic intensities, reducing the sensitivity of the metric. Incorporating co-temporal SDO/HMI magnetograms in future work would enable a physically motivated intensity–flux analysis expected to follow a power-law relationship [4].

QUESTION 3 ANSWERED: Segmentation Method Comparison

VII. QUESTION 3: SEGMENTATION METHOD COMPARISON

A. Segmentation Results

Table VI summarises region counts produced by each method across all four SUIT narrowband filters for the 07 January 2024 observation. NOAA Solar Region Summary data for that date catalogued three active regions: AR3536, AR3539, and AR3540.

TABLE VI

Detected Region Counts by Segmentation Method and Filter

Filter	Fixed Threshold	Otsu Threshold	Adaptive Threshold
NB04	6	1	106
NB05	3	1	101
NB06	1	1	106
NB07	2	1	97

B. Segmentation Visualisations

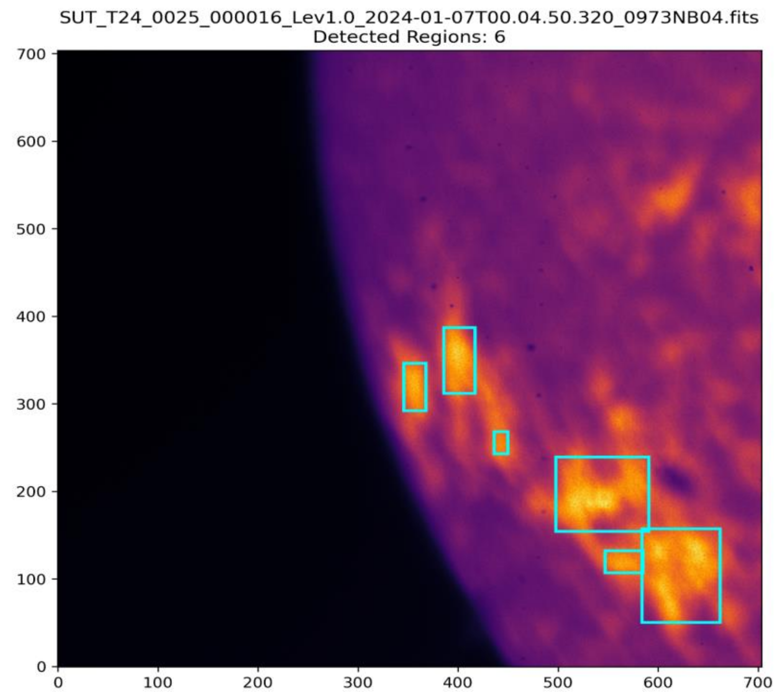

Figure 3. Fixed-threshold segmentation result for SUIT NB05, 07 January 2024. Detected active regions are enclosed by bounding boxes.

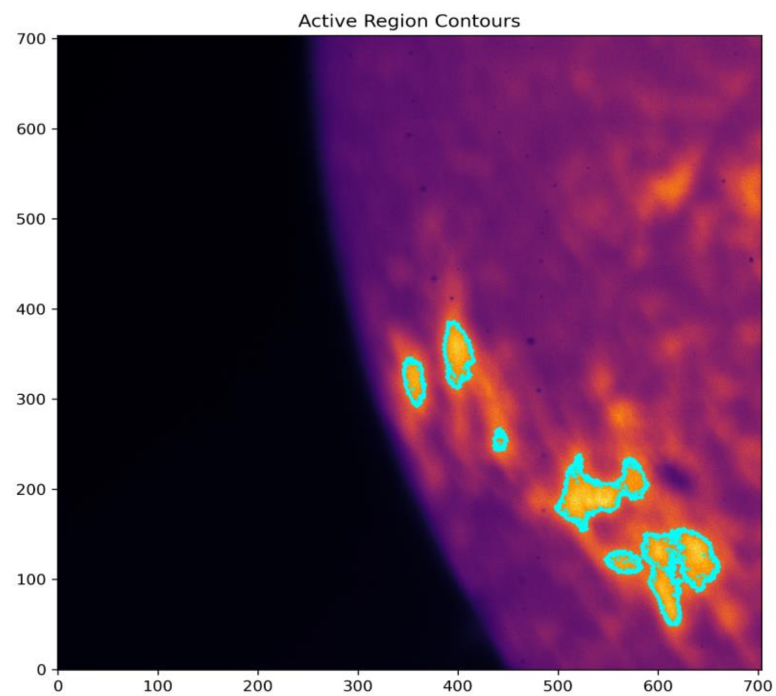

Figure 4. Contour overlay for active regions detected in the NB04 filter image.

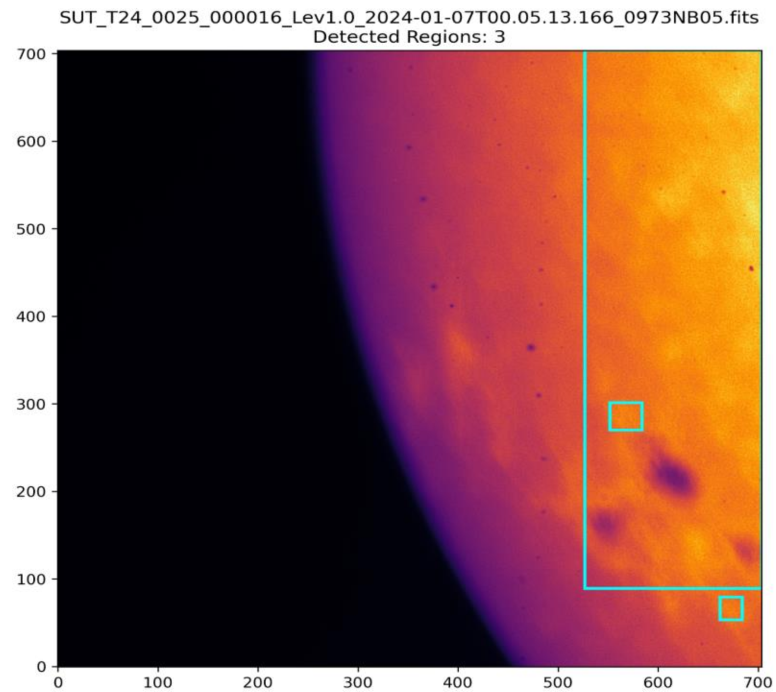

Figure 5. Fixed-threshold detection result for NB05 showing three regions — consistent with NOAA observations.

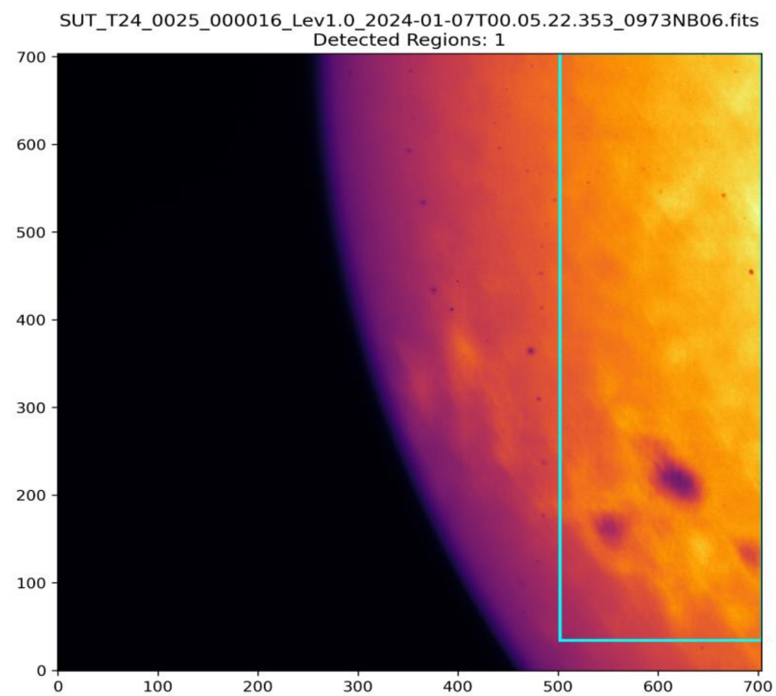

Figure 6. Fixed-threshold detection result for NB06 showing one dominant active-region structure.

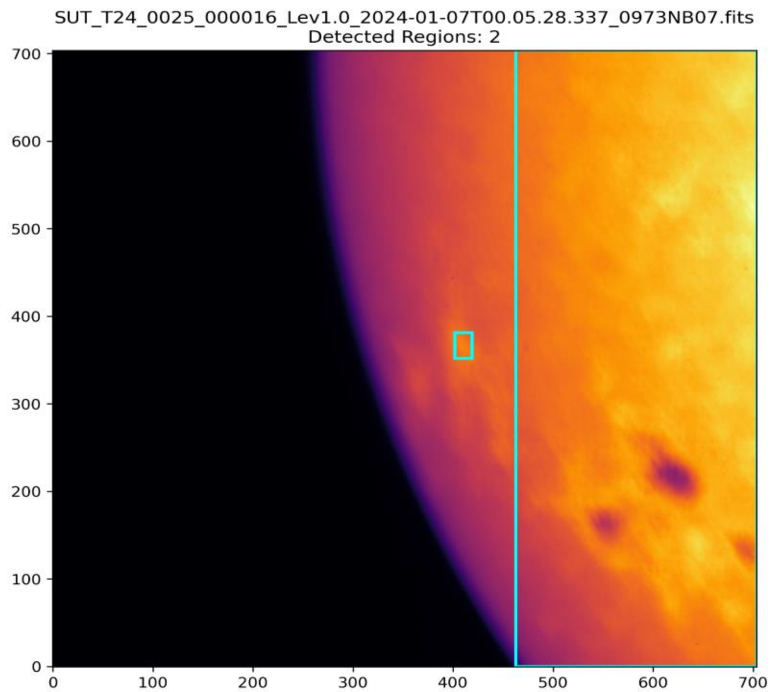

Figure 7. Fixed-threshold detection result for NB07 showing two detected active regions.

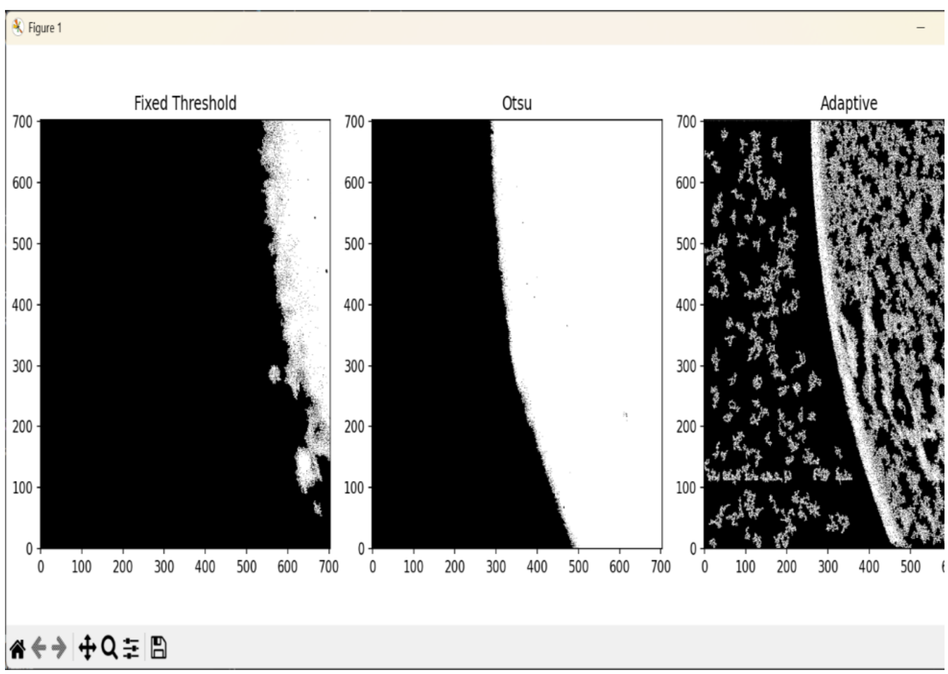

Figure 8. Side-by-side comparison of fixed, Otsu, and adaptive thresholding applied to SUIT NB05, 07 January 2024.

C. NOAA Comparison and Discussion

Fixed thresholding produced a region count of three in the NB05 filter, exactly matching the three NOAA-catalogued active regions for 07 January 2024. The method generated spatially coherent contours enclosing compact, physically plausible regions consistent with sunspot-group morphology.

Otsu thresholding produced a uniform count of one region across all filters. The algorithm merged the three spatially separated active regions into a single connected structure because the Otsu threshold converged well below 0.7, including bright quiet-Sun areas in the mask. Adaptive thresholding produced counts of 97–106 regions per filter by classifying chromospheric network elements, limb-brightening artefacts, and instrument noise as active-region candidates, resulting in severe over-segmentation that would require extensive post-processing to use practically.

Fixed thresholding therefore achieves the best balance of sensitivity and specificity for routine SUIT active-region monitoring. Otsu thresholding may be useful when only a coarse binary mask of enhanced emission is required, while adaptive thresholding would benefit from a large minimum-area filter or a morphological post-processing step before deployment.

VIII. OVERALL DISCUSSION

The three analyses collectively characterise solar active-region behaviour from complementary perspectives. The temporal study confirms that SUIT NB05 observations reliably capture growth and decay dynamics over periods of days to weeks, with measured area variations broadly consistent with Solar Cycle 25 activity records. The correlation analysis, while statistically inconclusive at the current sample size, highlights the data requirements for future work: co-temporal magnetogram coverage is essential for any investigation of the physical relationship between chromospheric ultraviolet emission and the underlying magnetic field. The segmentation comparison provides clear practical guidance: fixed thresholding remains the primary detection method, while Otsu and adaptive methods may serve as components of more sophisticated multi-stage pipelines or ensemble classifiers.

IX. LIMITATIONS

1. Only five observation dates were analysed; robust temporal characterisation requires several dozen observations over one or more solar rotation periods.
2. Co-temporal SDO/HMI magnetogram data were unavailable, preventing direct intensity–magnetic flux correlation.
3. The fixed threshold of 0.7 may not generalise to observations acquired during different phases of the solar cycle.
4. Region-count-based NOAA validation does not confirm spatial correspondence between detected and catalogued regions.
5. Adaptive thresholding parameters were not optimised through cross-validation.

X. FUTURE WORK

1. Extend the temporal dataset to cover a full solar rotation (≈27 days) at daily cadence to enable rotation-tracking of active-region complexes.
2. Integrate co-temporal SDO/HMI line-of-sight magnetograms for intensity–magnetic flux correlation analysis.
3. Develop a U-Net deep-learning segmentation model trained on annotated SUIT observations to improve detection precision.
4. Compute spatial validation metrics (IoU, precision, recall) by co-registering detections with NOAA heliographic coordinates.
5. Investigate multi-filter fusion strategies combining detections from two or more SUIT narrowband channels.

XI. CONCLUSION

This paper presented a three-question analysis of solar active regions using SUIT ultraviolet observations from Aditya-L1. Temporal analysis across five NB05 observations (January–February 2024) revealed active-region area ranging from zero to 155,340 pixels, capturing emergence and decay dynamics consistent with Solar Cycle 25 behaviour. Intensity correlation analysis yielded Pearson and Spearman coefficients of −0.13 and 0.00, indicating no statistically significant area–intensity relationship at the current sample size. Segmentation comparison demonstrated that fixed thresholding at 0.7 normalised intensity produces region counts exactly matching NOAA observations, outperforming Otsu thresholding (which under-segmented) and adaptive thresholding (which severely over-segmented). Together with Part 1, these results establish a validated, reproducible pipeline for solar active-region monitoring from SUIT data and provide a quantitative foundation for future machine-learning-based space-weather applications.

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